

Assignment IV - PHYS 708

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Problem 1

(a) The propagation of cosmic-ray primary species from a uniform distribution of sources in the Galactic disk follows

$$\nabla \cdot (D_p \nabla N_p) - 2h n v_p \sigma_p \delta(z) N_p = -S_p,$$

where $S_p(\mathbf{r}, E) = \mathfrak{R} Q_p(E) \delta(z)$ is the source term assumed to be extended up to the radial distance R . Using the solution of the above equation given on Slide 55 of Lecture 8, show that the spectrum in the Galaxy at very high energies follows a power law of the form $N_p(E) \propto E^{-(\gamma+\alpha)}$ if the source spectrum and the diffusion coefficient follows $Q_p(E) \propto E^{-\gamma}$ and $D_p(E) \propto E^\alpha$ respectively. [Hint: Refer to Slides 54-55 of Lecture 8].

(b) Under the same model given above, show that the spectrum of cosmic-ray secondaries at very high energies in the Galaxy follows $N_s(E) \propto E^{-(\gamma+2\alpha)}$.

(c) Also show that the secondary-to-primary ratio follows an energy dependence of $\frac{N_s}{N_p} \propto E^\alpha$. [Hint: Refer to Slide 58 of Lecture 8].

Variables in Study

$N_p(\mathbf{r}, z, E)$	Primary CR density (function of position and energy)
$N_s(\mathbf{r}, z, E)$	Secondary CR density
$Q_p(E)$	Primary source spectrum; on slides: $Q_p(E) \propto E^{-\gamma}$
$D(E)$	Spatial diffusion coefficient; on slides: $D(E) \propto E^\alpha$
$D_p(E), D_s(E)$	Diffusion coefficients for primaries and secondaries (same scaling E^α)
n	Gas density in the thin disk
h	Half-thickness of the source disk (so thickness $2h$), with $\delta(z)$ on the slides
$v \simeq c$	Particle speed (appears with n and cross sections on the slides)
σ_p	Inelastic destruction cross section for primaries
σ_{ps}	Production cross section for secondaries from primaries
σ_s	Total destruction (loss) cross section for secondaries
H	Halo half-height; vertical boundary at $z = \pm H$
R	Radial extent of sources (no radial boundary for CRs; appears inside the Bessel integral)
K	Radial wavenumber in the integral representation
J_1	Bessel function of the first order (order 1)
$\mathcal{R}$	Supernova explosion rate factor used on the slides

Solutions to the Transport Equation for Primary & Secondary Cosmic Rays

Primaries (Slide 54/55). With the transport equation

$$\nabla \cdot (D_p \nabla N_p) - 2h\nu\sigma_p \delta(z) N_p = -S_p, \quad S_p(r, E) = \mathcal{R} Q_p(E) \delta(z),$$

the solution is

$$N_p(z, E) = \frac{R \mathcal{R} Q_p(E)}{2D_p(E)} \int_0^\infty \frac{\sinh[K(H-z)]}{\sinh(KH) \left[K \coth(KH) + \frac{2hnc \sigma_p}{2D_p(E)} \right]} J_1(KR) dK. \quad (1)$$

Secondaries (Slide 58). With the source $S_s(r, E) = 2hnc \sigma_{ps} \delta(z) N_p(r, E)$ and

$$\nabla \cdot (D_s \nabla N_s) - 2h\nu\sigma_s \delta(z) N_s = -S_s,$$

the slide gives

$$N_s(z, E) = 2hnc \sigma_{ps} N_p(z, E) \frac{R}{2D_s(E)} \int_0^\infty \frac{\sinh[K(H-z)]}{\sinh(KH) \left[K \coth(KH) + \frac{2hnc \sigma_s}{2D_s(E)} \right]} J_1(KR) dK. \quad (2)$$

Solution

All results below refer to the *very high energy* regime of cosmic rays. We explore the power law spectra for primary & secondary cosmic rays by studying the energy dependencies in their solutions.

(a) Primary spectrum $N_p(E)$ at very high energies

From (1), the energy dependence arise from factor $\frac{Q_p(E)}{D_p(E)}$ outside the integral ; inside the integral there is only E -dependence containing the $D_p(E)$: $\frac{2hnc \sigma_p}{2D_p(E)}$. At very high energies $D_p(E) \propto E^\alpha$ becomes large, so

$$\frac{2hnc \sigma_p}{2D_p(E)} \ll K \coth(KH),$$

and the integral becomes effectively *energy independent*. Therefore

$$N_p(E) \propto \frac{Q_p(E)}{D_p(E)} \propto \frac{E^{-\gamma}}{E^\alpha} = \boxed{E^{-(\gamma+\alpha)}}.$$

(b) Secondary spectrum $N_s(E)$ at very high energies

From (2),

$$N_s(E) \propto [N_p(E)] \times \frac{1}{D_s(E)} \times \left(\text{integral that is energy independent at high } E \right).$$

Using the result of part (a), $N_p(E) \propto E^{-(\gamma+\alpha)}$, and $D_s(E) \propto E^\alpha$ (same rigidity scaling on the slides), we get

$$N_s(E) \propto E^{-(\gamma+\alpha)} E^{-\alpha} = \boxed{E^{-(\gamma+2\alpha)}}.$$

(c) Secondary-to-primary ratio

Taking the ratio using the high-energy limits above,

$$\frac{N_s}{N_p} \propto \frac{1}{D_s(E)} \propto \boxed{E^{-\alpha}}.$$

As noted on Slide 58, this ratio is independent of the source parameters at very high energies.

Problem 2

For a time-dependent propagation of cosmic-ray primaries from a nearby source, the transport equation is given by

$$\nabla \cdot (D_p \nabla N_p) - 2h n v_p \sigma_p \delta(z) N_p + Q_p = \frac{\partial N_p}{\partial t},$$

where $Q_p(r, E, t) = Q_p(E) \delta(r) \delta(t - t_0)$ is the source term injecting cosmic rays in a burst-like way. All other parameters have the same definition as in Question (1). Show that the spectrum at very high energies follows $N_p(E) \propto E^{-(\gamma+3\alpha/2)}$. [Hint: Refer to Slide 70 of Lecture 8].

Variables in Study

$N_p(\mathbf{r}, z, E, t)$	Differential number density of primary cosmic rays (CRs).
$Q_p(\mathbf{r}, E, t) = Q_p(E) \delta(\mathbf{r}) \delta(t - t_0)$	Burst-like source at time t_0 located at the origin.
$Q_p(E) \propto E^{-\gamma}$	Source spectrum of primaries (slide 70 notation).
$D_p(E) \propto E^\alpha$	Spatial diffusion coefficient (same rigidity scaling as slides 54–55).
n	Gas density in the disk.
σ_p	Inelastic destruction cross section for primaries.
$v \simeq c$	Particle speed (appears with n and cross sections on the slides).
h	Half-thickness of the gas disk (thin-disk loss term uses $\delta(z)$).
r	3D distance to the source.
z	Vertical coordinate.
$\Delta t \equiv t - t_0 > 0$	Time since the burst.
$q = z / \sqrt{D_p(E)}$	Auxiliary variable on slide 70.
$b = \frac{2h n c \sigma_p}{2\sqrt{D_p(E)}}$	Auxiliary coefficient on slide 70.

Solution to the Transport Equation for a Discrete Nearby Source

The time-dependent transport equation used on slide 70 is

$$\nabla \cdot (D_p \nabla N_p) - 2h n v \sigma_p \delta(z) N_p + Q_p = \frac{\partial N_p}{\partial t}.$$

Using the Green-function result shown on slide 70, the unbounded solution for a burst source is

$$N_p(r, E, t) = \frac{Q_p(E)}{8\pi D_p(E)^{3/2} \Delta t} \exp\left[-\frac{r^2}{4D_p(E) \Delta t}\right] \times \left\{ \frac{e^{-q^2/[4\Delta t]}}{\sqrt{\pi \Delta t}} - b e^{bq+b^2\Delta t} \operatorname{erfc}\left(b\sqrt{\Delta t} + \frac{q}{2\sqrt{\Delta t}}\right) \right\}, \quad (3)$$

with q and b as defined above.

Solution to the problem (high-energy spectrum)

We are asked to determine the energy dependence of $N_p(E)$ at very high energies. We use $D_p(E) \propto E^\alpha$ and $Q_p(E) \propto E^{-\gamma}$ as given in the question.

High-energy limit When E is large, $D_p(E)$ is large. Then:

- $q = |z|/\sqrt{D_p} \rightarrow 0$, $b \propto 1/\sqrt{D_p} \rightarrow 0$ and $\operatorname{erfc}\left(b\sqrt{\Delta t} + \frac{q}{2\sqrt{\Delta t}}\right) \rightarrow 1$, so the curly bracket in (3) approaches a factor that depends on Δt but is *independent of E* .
- The Gaussian factor $\exp[-r^2/(4D_p\Delta t)] \rightarrow 1$ for large D_p (no E -dependence).

Therefore, at high energies the only E -dependence comes from the explicit prefactor in (3):

$$N_p(E) \propto \frac{Q_p(E)}{D_p(E)^{3/2}} \propto \frac{E^{-\gamma}}{E^{3\alpha/2}} = \boxed{E^{-(\gamma+\frac{3}{2}\alpha)}}.$$